

Human Exposure to Mobile Phone EM Radiation

The rapid development of telecommunication technologies has resulted in an increase in the human exposure to electromagnetic fields and led to a number of scientific studies exploring its possible health hazards. This article highlights how a mobile phone creates radiation fields around the human head resulting in harmful health effects



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In many countries, over half the population uses mobile phones and the market is growing rapidly. Mobile phones emit electromagnetic radiation in the microwave range, and exposure to these electromagnetic (EM) fields emitted by the mobile phones has greater likelihood of causing adverse health effects.

Unlike X-rays and gamma rays, radiation from microwave telecommunication systems falls under non-ionising type that cannot cause ionisation in the human body. All radiations and fields of the electromagnetic spectrum that do not normally have sufficient

energy to produce ionisation in matter characterised by energy per photon as less than about 12 electron volts, wavelengths greater than 100 nanometres and frequencies lower than 3×10^{15} Hz are termed as non-ionising.

For broadcasting purpose, high RF power is generally required to maximise the area of coverage. Electric field strengths in the vicinity of antennae can reach several hundred volts per metre. Even higher values can be found close to occupational sources. In many such applications RF safety problems arise as RF power is high.

Sources used by the general public, e.g., for wireless communication or data transmission, generate compara-

bly much lower fields near the position of the user. But this may also depend on the behaviour of the user, especially concerning the distance to the source.

Cellular mobile communication networks, on an average, cause low levels of electromagnetic fields in areas accessible to the general public. Handsets and cell phones, however, might cause significantly higher peak levels of exposure during use.

Mobile phones are low-powered radio-frequency transceivers, operating at frequencies between 450 and 2700 MHz. Typically, a GSM (global system for mobile communication) handset emits radio waves having peak power of 2 W, and CDMA2000 (code-division multiple access) handset emits below 1W peak power. The handset transmits power only when it is turned on.

In most systems, the cellphone and the base transceiver station check the reception quality and signal strength, and the power level is increased or decreased automatically, within a certain span, to accommodate different situations such as inside or outside buildings and vehicles. The power (and hence the radio-frequency exposure to a user) falls off rapidly with increase in the distance from the handset. A person using a mobile phone 30-40 cm away from his body—for example, when text messaging, accessing the Internet or using a hands-free device—will therefore have a much lower exposure to radio-frequency fields than someone holding the handset against his head.